Figure 1: Purly the Snowman

At the time of starting this assignment, it is assumed that you have already completed assignments 1 and 2. That means you already have your “Purly” character that can move on the game surface and can rotate 360 degrees on its feet (Y-axis). In addition to this, Purly can pop the balloons placed on the walls by reaching them. At the same time, it needs to dodge the attacking balls from the two snowguns. The lifetime for each attacking snowball is 10 seconds, and it will be thrown by snowguns towards ‘Purly’ every 20 seconds.

In this assignment 3, you will apply some modifications to the current version of the game. The two snowguns **will not** throw the attack balls **at the same time**. Rather, they will operate individually (i.e. one gun at a time). The operating time gap between two guns will be 20 seconds. One snowgun will be activated at a time. In addition, unlike last version, the attacking balls will not follow the “Purly”, rather, the attacking snowballs will follow a straight direction towards the **opposite wall**, but this direction will be random. And this time the attacking speed of the snowball will be at least 6.5f.

Moreover, the balloons on the walls will not be fixed this time. The balloons will be placed randomly on four walls. Every time, one balloon is popped on one wall, and within the next two seconds, another balloon will be placed in **a random position** on that wall. By this, all the time in the game board, there will be four balloons present.

Next, in this version of the game, you need to implement a score-calculation mechanism. For each balloon Purly pop will be calculated as 1 point. This point calculation will continue until Purly dies. And Purly dies only when it gets hit by an attacking snowball. The points should be stored and tracked for the future use in a local file. At the same time, the live points need to be displayed

in the game-playing environment. In addition to this point calculation, an additional menu system needs to be present. You need to implement three menu buttons in this menu system: i. pause menu, ii. Restart menu and iii. Resume menu. At the beginning of the game, the resume menu will remain in-active, and pause menu and restart button will remain active. During the game is paused, the main menu will be shown (discuss later). Please see figure 2 to understand the game scenario in detail:

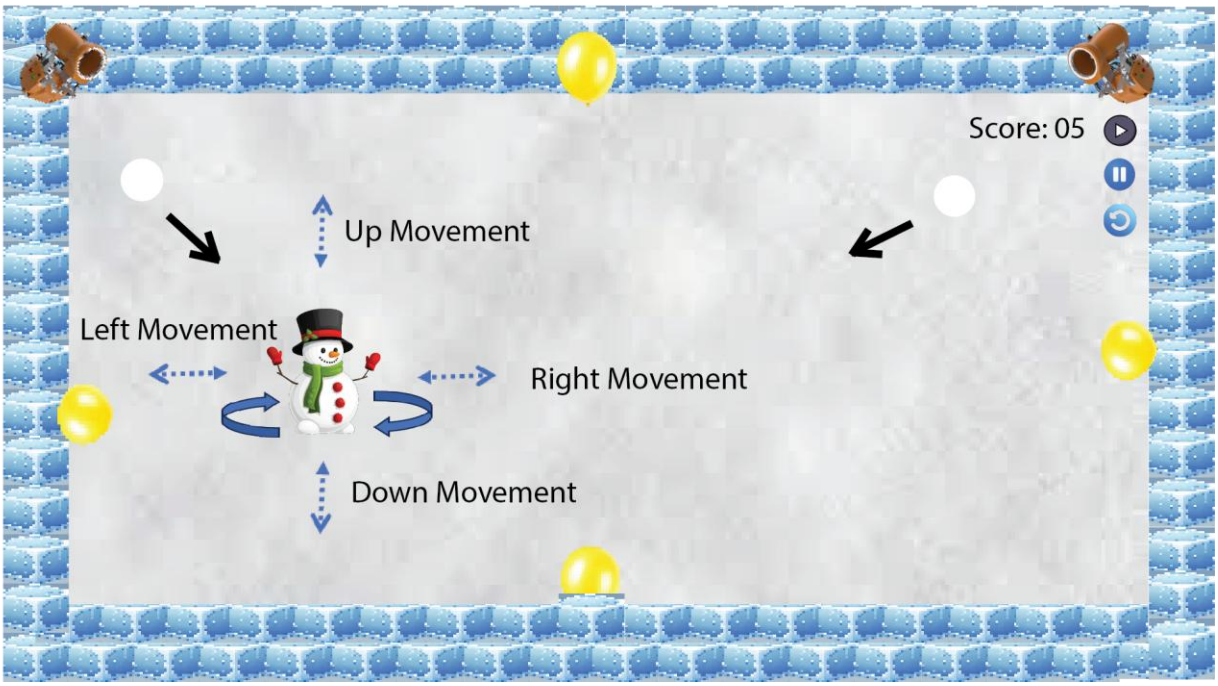

Figure 2: Game Arena

In this assignment, the game will have a landing page/starting page. Once you start the game, this page will be shown. The landing page should look like near to figure 3. We can call this page the menu page. The menu page will have the following menu options: i. New Game, ii. Save game, iii. High Scores, iv. End game. Among these, the new game menu will start the game, save game will resume the last paused game, End game will close the game. The high scores menu will list the last five scores with the player name in a textbox/(in a new page with game UI layouts). In addition to these menu options, an “enter player name” UI should be present to take the name from the player. An example of a starting page is given in Figure 3.

Moreover, in this assignment, you need to use the new Input System, not the old input manager. A lot of you have already implemented Input System in the assignment but not been evaluated yet. In this assignment you are going to be evaluated based on the input system, its working conditions and proper activities. All the movements of “Purly” in this assignment needs to be controlled by Input System. The Input System should have separate movement type for walking and rotations. If you use only one type for movement, then you will loose 50% points for this section as stated in the rubric.

In addition, If you can list the high scores with player names in descending order (higher to lower) you will be awarded 5% more points.

Figure 3: Loading Page

In this assignment you are going to do the following steps and will be evaluated based on the related evaluation marks:

	Assignment Steps	Marks
a.	Update the active period of the snowguns. Make sure two snowguns are not going to work same time, and there remains gap in between their active time.	5
b.	Snowballs will follow a straight line with faster speed from gun point toward other wall, but the direction will be random. Make sure they are not following the Purly any more.	10
c.	The balloons on the four walls will continuously generate every time after popped. But they will be placed randomly on different positions on the walls. Moreover, once one balloon is popped, there will be 2 2-second time gap to generate a new one.	10
d.	A score calculation mechanism needs to be present in your game. And the live scores need to be displayed on the game scene.	10
e.	A pause menu system with provided directions need to implemented properly and need to active. Missing any conditions/instructions will result 0 point for this section.	15

f.	Game staring/Landing page needs to be added with proper active menu buttons.	15
g.	Player name registration and saving scores related to the player name need to be present and working properly.	10
h.	List the player names with scores under the history button. You can either use textbox or a new game scene with UI tools.	5
i.	Once attacking snow ball touch Purly, it will die	5
j.	Player movement needs to be reimplemented using Input System (MVC framework based).	10
k.	Sort the player lists with score based descending order and display them.	5
	Total	100

Submission Rules:

1. Omnivox/LEA might not allow you to upload all the game files. Your Google Drive/College provided one drive to store your summative assignment 03 and share the link with me. You can also upload your summative assignment in Github as well.
2. Before final submission, make a **screen recording** with all the implemented active features as instructed in this assignment. Please submit this recording with your assignment 03 in the LEA.
3. Check off the assignment with your **teacher in-person in class**. Otherwise, you will get 0 for the submission.
4. Make sure your application is active and executing properly when you are going to check-off with teacher. And make sure you checked-off with the teacher within 1 week from the regular submission deadline. If you fail, it will automatically be a 0, even though you submitted the work and recordings in time.